Semantic Affinity-Driven Spatiotemporal Transformer Network for Satellite Video Moving-Object Segmentation

Yixuan Lv, Zhuang Zhou, Kang Liu, Han Wang, Jian Yang, Shengyang Li, Jianing You, and Kailun Zhang

***Abstract*— Satellite video intelligent processing plays a critical role in Earth observation applications such as traffic monitoring and environmental surveillance. However, moving-object segmentation in satellite videos faces several challenges. First, spatiotemporal redundancy makes it difficult to model long-range dependencies because large-scale scenes with slow background changes lead to fragmented segmentation. Second, semantic ambiguity arises when stationary objects like parked aircraft share category-level similarities with moving targets, which causes false positives. Besides, insufficient feature discrimination occurs as small, rigid objects such as ships exhibit weak texture and edge details under low-resolution imaging. To overcome these issues, we introduce a semantic affinity-driven spatiotemporal Transformer network that leverages a Transformer-based architecture to capture pixel-level dependencies across spatial and temporal dimensions. Furthermore, our network employs a contextual affinity-constrained decoder to suppress category-level interference and integrates a triple-branch feature extractor with edge priors for enhanced contour delineation. Our framework operates in an end-to-end manner without requiring fine-tuning during inference, which ensures deployment efficiency. Extensive experiments on a dataset built upon SAT-MTB demonstrate state-of-the-art performance with a J&F Mean of 71.7%. The proposed method outperforms the baseline by 3.7% with improvements of 4.4% in J-Mean and 3.1% in F-Mean. In addition, it surpasses the optimized SAM2 with a 10.7% higher J-Mean while maintaining a significantly smaller parameter count (34.6 M versus 224 M). Both qualitative and quantitative evaluations confirm the method's superiority and temporal stability. This work offers a robust and efficient solution for accurate moving-object segmentation in satellite videos.**

***Index Terms*— Affinity, deep learning, moving-object segmentation, satellite video, Transformer.**

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I. Introduction

VIDEO satellites represent a novel class of Earth observation satellites, offering continuous "staring" capabilities over specific regions to capture high-temporal-resolution images. Compared with traditional Earth observation satellites, video satellites provide unprecedented spatiotemporal data streams that enable detailed monitoring and analysis of dynamic phenomena on Earth's surface. Recent advancements in Earth observation satellites, including the launch of SkySat-1 in 2013 (the world's first submeter video satellite) [1], have provided new capabilities for high-frame-rate imaging. The Changguang Satellite Constellation, which consists of 138 satellites equipped with dedicated video imaging capabilities, enables the continuous acquisition of high-resolution video sequences at 30 frames/s with high spatial and spectral fidelity. This evolution in Earth observation technology has significantly enhanced our ability to monitor dynamic processes on Earth. The integration of video imaging capabilities into modern satellite constellations represents a transformative leap in remote sensing, paving the way for more comprehensive and timely insights into environmental changes and human activities.

Intelligent processing and analysis of satellite video can automatically extract and analyze information for scenes of interest. It will provide important technical support and play an important role in the above practical application scenarios. Satellite video object detection, segmentation, and tracking are several typical tasks. Among them, satellite video object segmentation (VOS) can achieve refined segmentation of targets of interest in time-continuous scenes. Compared with detection and tracking, segmentation can obtain additional refined description information while locating the object position, which can serve for further refined analysis and applications.

Intelligent processing and analysis of satellite video enables automatic extraction and interpretation of scenes of interest, providing technical support for various practical applications. Video object detection, tracking, and segmentation are representative tasks in this field. In particular, VOS achieves fine-grained, temporally consistent target delineation, offering richer structural and contextual information than detection or tracking alone, which facilitates more precise downstream analysis.

Satellite VOS presents several unique challenges. First, unlike natural scene videos, satellite videos are captured from

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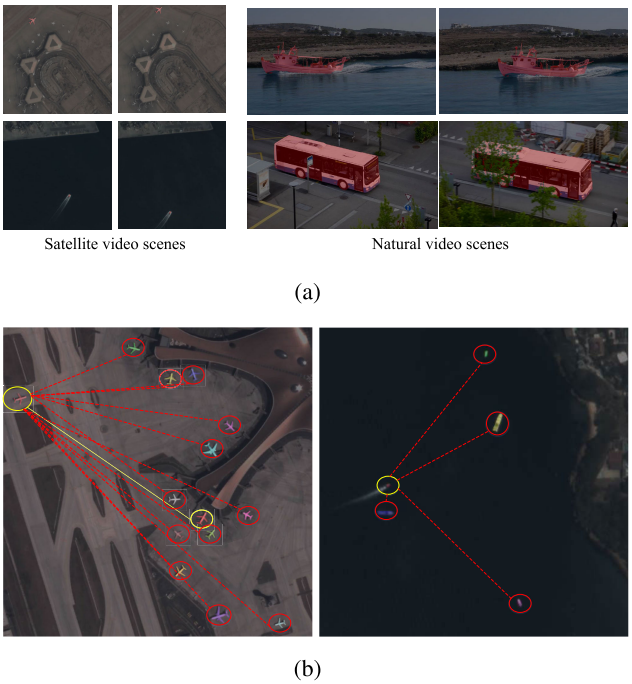


Fig. 1. Challenges in the field of satellite video. (a) Scene comparison. (b) Interference caused by the semantic similarity between the foreground and the background.

high altitudes with a nadir view and typically have a spatial resolution of about 1 m—lower than that of conventional optical imagery or high-resolution remote sensing data. Objects of interest, such as vehicles, ships, and aircraft, often appear small with limited texture and shape details, resulting in low contrast and poor distinguishability. Moreover, the high frame rate and long imaging distance lead to minimal interframe motion and slow background variation, introducing significant spatiotemporal redundancy. As shown in Fig. 1(a), objects occupy only a small portion of large-scale scenes, making fine-grained details difficult to perceive. Second, temporal information is a key attribute of satellite video data. Unlike static images, satellite videos allow for dynamic analysis over time; however, many applications focus only on moving targets, while most foreground and background objects remain stationary and visually similar. This semantic similarity between target and nontarget regions can cause interference, especially in complex scenes such as airports and harbors, as illustrated in Fig. 1(b). Third, effectively modeling both inter- and intra-frame dependencies remains challenging. Temporal relationships are essential for capturing consistent object features across frames, particularly for tracking dynamic targets. Meanwhile, spatial relationships within individual frames provide critical appearance and structural cues for accurate mask generation despite intraclass similarity and appearance variations. As shown in Fig. 2, methods relying solely on spatial learning (e.g., convolutional neural network (CNN)-based approaches) often fail to exploit temporal dynamics and long-range semantic dependencies, limiting their performance on large-scale satellite video sequences.

For satellite VOS tasks, the existing methods are mainly divided into unsupervised and semi-supervised. Conventional unsupervised satellite VOS methods mainly utilize the idea of foreground extraction. Commonly used motion foreground

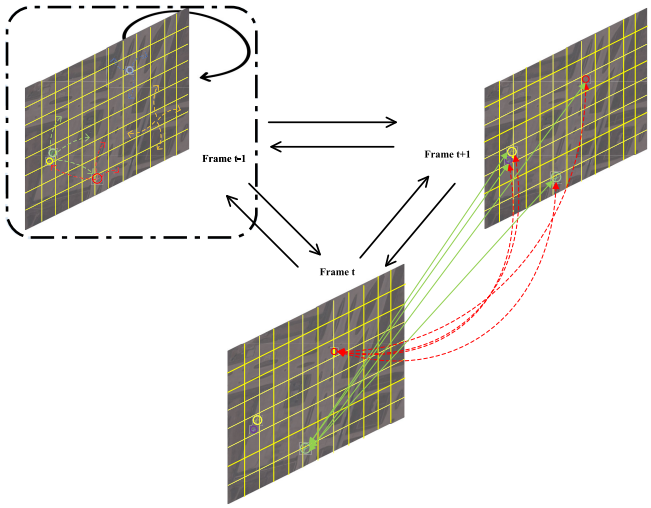


Fig. 2. Overview of the proposed method for comprehensive spatiotemporal semantic learning at the pixel level.

detection and extraction methods are optical flow models, background subtraction models, frame difference models, and visual background extractor models. Zhang et al. [2] integrated low-rank matrix decomposition with structured sparsity regularization to achieve moving-object extraction. They also proposed an online low-rank and structured sparse decomposition to mitigate batch processing delays [3]. Shi et al. [4] designed a normalized frame difference labeling algorithm to detect moving airplanes. However, these traditional foreground extraction models have high requirements for video quality and consider the background to be static. Satellite video, due to the movement of its shooting platform and the change of viewpoints, the background usually has slow-motion changes as well. Therefore, these foreground extraction models typically extract many background shadows and false positives. Moreover, the above algorithms require a large amount of human effort and have a single feature expression, making the processing more complicated.

In recent years, rapid advancements in deep learning and computer vision have significantly impacted VOS. CNNs, with their strong feature representation capabilities, have enabled a wide range of VOS methods to be developed and applied. These approaches typically incorporate varying degrees of prior knowledge as input. Some methods process each frame independently without exploiting temporal information across frames. For instance, OSVOS [5], a classical one-shot VOS framework, applies a CNN trained offline and fine-tuned online for single-frame segmentation. While effective on static frames, its performance degrades when temporal coherence is required due to the lack of interframe modeling. Other approaches leverage temporal dependencies by incorporating the segmentation results from previous frames. MaskTrack [6] employs a semantic segmentation network that uses the prediction from the preceding frame as an initial estimate for the current frame. However, it heavily relies on prior predictions, leading to error propagation if earlier frames are inaccurately segmented. To improve robustness, Khoreva et al. [7] introduced additional motion cues via optical flow amplitude maps between consecutive frames. This dual-guidance strategy

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enhances accuracy but increases computational cost due to CNN-based optical flow estimation. In satellite video segmentation, these methods often overlook the semantic interference caused by stationary background objects. Zhong et al. [8] proposed a spatiotemporal dual-branch network based on GANs to model the mapping from reference sequences to query frames. However, their method focuses on global image-level relations and neglects pixel-level spatiotemporal dependencies, requiring further fine-tuning during inference. More recently, Kou et al. [9] introduced a global–local matching strategy to learn cross-temporal relationships between reference and query frames for satellite video segmentation. Nevertheless, this approach does not consider temporal consistency among reference frames and incurs a high computational cost through CNN-based matching mechanisms.

To address the above challenges, we propose an end-to-end semantic affinity-driven spatiotemporal Transformer that effectively models both inter- and intra-frame pixel-level dependencies. First, to exploit the spatial and temporal redundancy inherent in satellite video sequences, our framework employs a Transformer architecture that captures long-range spatiotemporal relationships through self-attention mechanisms across all input elements (as shown in Fig. 2. Second, in single-object segmentation tasks, stationary background objects with similar appearance often cause semantic interference. To mitigate this, we introduce contextual semantic affinity constraints that learn discriminative feature associations to enhance target–background separability. Finally, considering the small size and indistinct contours of typical rigid targets such as ships and aircraft in satellite videos, we further incorporate an edge feature enhancement module, which exploits shape priors to improve boundary delineation.

In all, the contributions of this article are summarized as follows.

1) We introduce an end-to-end semantic affinity-driven spatiotemporal Transformer network designed for moving-object segmentation in satellite videos that operates effectively without the need for fine-tuning during inference. By leveraging both explicit supervision and implicit pixel-level spatiotemporal modeling to learn stable object representations and capture precise spatial structures. This dual strategy significantly enhances mask generation accuracy.
2) To mitigate the semantic interference arising from the similarity between foreground and background elements, we develop a contextual semantic affinity-constrained mask decoding. This design facilitates category-aware, fine-grained semantic robustness, which significantly improves segmentation performance in complex scenes.
3) Recognizing that targets such as aircraft and ships in satellite videos are typically rigid and exhibit nondeformable contours, we propose a three-branch input layer that integrates a shape prior into the feature extraction process. This module specifically enhances edge features, leading to improved delineation of small and subtle objects.

We conduct experiments comparing our method against traditional unsupervised approaches, deep learning-based techniques, and the visual foundation model. The results demonstrate the superior performance of our approach. The rest of this article is organized as follows. Section II reviews related works and research on VOS. In Section III, we detail our proposed method. Section IV describes the experimental setup, evaluation metrics, and presents both qualitative and quantitative results. Finally, Section V concludes the article and outlines directions for future work.

II. Related Work

VOS is a binary labeling problem that aims at recognizing and describing one or more objects appearing in a video. Video object recognition and segmentation are the basis of scene understanding and are important for research in some fields due to their accurate labeling down to the pixel level and temporal–spatial correlation in segmentation results.

In recent years, deep learning has been successful in tasks such as image recognition, object detection, and instance segmentation [10], [11], [12]. However, these tasks are for still images, and VOS is a more complex problem than these tasks, and it is considered an extension of these classical tasks. These algorithms in still images do not simply extend to VOS. Compared to object tracking, VOS faces similarly diverse scenarios, while the tracking task yields a bounding box enclosing the object, segmentation is a much more delicate task of correctly categorizing pixels belonging to the object in the video and separating them from the background.

Depending on the degree and manner of human involvement, VOS methods are generally categorized into unsupervised VOS, semi-supervised VOS, and interactive VOS. However, in practical application scenarios, interactive segmentation methods are more limited due to their reliance on manual intervention. This article primarily focuses on unsupervised and semi-supervised methods. Specifically, unsupervised VOS does not require any user input during the testing phase and usually automatically segments the most critical, salient objects in the video. On the other hand, for semi-supervised VOS, the user typically provides the object mask of the first frame or a keyframe, after which the method proceeds to segment the object in subsequent frames.

A. Unsupervised Video Segmentation

Early VOS methods were mainly based on geometry and limited to a specific motion context. The idea is to model the background appearance of each pixel and consider rapidly changing pixels as the foreground, with any significant change in the image and background model representing a moving object, and use the connectivity component algorithm to estimate the connectivity region corresponding to the object. The hybrid Gaussian distribution (GMM)-based background–foreground segmentation algorithm is a classical algorithm that counts the pixel values at any coordinate on a video frame in the time direction and assigns a number of Gaussian probability density functions to each coordinate as a model for the probability distribution of the pixel values at that location [13].

Some methods first generate point trajectories, then use the matching matrix of the trajectories to cluster the trajectories,

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and finally use the clustered trajectories as a priori information to obtain video target segmentation results [14], [15]. Some methods achieve saliency detection based on low-level features such as texture, color, and so on and further check the motion information of the object to achieve VOS [16], [17]. Papazoglou et al. [18] utilized motion boundary information to achieve fast and fully automated VOS. However, due to limitations in the ability to characterize, these methods are prone to fragmentation of the segmentation results and do not provide a complete representation of the object-level information.

With the help of the hot deep learning technology, there are some methods that combine deep learning to realize unsupervised VOS [19], [20], [21]. Early major VOS methods first generated saliency objects through complementary convolutional networks and then propagated video objects and hyperpixel-based neighborhood invertible streams through the video. Song et al. [22] extracted multiscale spatial features using a pyramid expansion (null) convolution structure, extracted different scale sequence features using a ConvLSTM module, and then concatenated output saliency maps via the full convolutional layer. Based on the obtained saliency probability map of the video sequence, the segmentation task is regarded as an energy function minimization problem. Based on the energy function of a fully connected conditional random field (CRF) model, the object segmentation is accomplished in the form of postprocessing. In conclusion, background elimination and point-trajectory methods simply assume that the object to be segmented in the video is the moving foreground, which tends to fragment the segmentation results and cannot express the object-level information in a complete and friendly way.

B. Semi-Supervised Video Segmentation

Early approaches typically address a number of spatiotemporal graphs with hand-designed features, including appearance, boundaries, and optical flow, and propagate foreground regions throughout the video. These approaches usually rely on two important cues: the target representation of the graph structure and the spatiotemporal connectivity relationship. Recently, a series of methods based on deep learning have emerged [5], [6], [23], [24], [25], [26], [27], [28], [29].

The first type of algorithm processes each frame of the video independently and does not involve interframe motion information [5], [23]. Caelles et al. [5] proposed the classical solution OSVOS, which utilizes a fully convolutional network trained offline and online on still images for single-frame VOS. The image classification network is first trained on ImageNet [30] to obtain the base pretrained model. They then continued training on the DAVIS dataset using a fully convolutional network, and at this step, the network had been able to separate foreground objects from background regions, but could not segment specific objects. Finally, an online segmentation model is trained for each video frame sequence, and the first frame of a given video sequence is fine-tuned online so that the network focuses on the foreground object. However, these methods do not utilize the temporal information of the video, while the online fine-tuning step is more time-consuming, which is not conducive to practical applications.

The second class of methods processes video frames by utilizing the segmentation result of the previous frame, so the processing of video frames is temporally correlated [6]. However, when this propagation-based idea encounters a situation where the objects of interest to be segmented gradually disappear completely in a video sequence and then gradually reappear, the effect is not optimal because there is no way to obtain the mask for the disappearance frame.

There are also matching-based approaches, where they first perform pixel-level matching between the features of the template frame and the current frame in the video and then segment each pixel of the current frame directly from the matching result [27], [28], [29]. Wang et al. [27] combined the ideas of propagation-based and matching-based and utilized the encoder structure of a Siamese network to obtain the template features of the first frame and the current frame features. They obtained the similarity feature maps through the correlation operation, and screened the foreground and background features through the ranking attention module. Then, they merged them with the predicted mask of the previous frame and obtained the final segmentation mask through the decoder structure.

C. VOS in Satellite Videos

Satellite video data provides rich temporal information that enables more refined object processing and analysis, making VOS a promising direction in Earth observation. Compared to traditional static remote sensing images, satellite videos offer time-continuous sequences, supporting motion modeling and semantic consistency across frames. However, objects of interest such as aircraft, vehicles, and ships often appear small, with blurred textures and low contrast against complex backgrounds, making accurate pixel-level segmentation particularly challenging.

Most existing methods adopt the semi-supervised VOS paradigm, where an initial mask from the first frame guides segmentation throughout the video sequence. Zhong et al. [8] proposed a spatiotemporal dual-stream network trained using the OSVOS framework, with spatial and temporal branches pretrained on PASCAL VOC 2012 [31] and ImageNet DAVIS [32], respectively. Their model was further fine-tuned on the DAVOS dataset, achieving improved region similarity and contour accuracy for aircraft targets.

Despite these advances, current approaches are limited in capturing long-range spatiotemporal dependencies and distinguishing moving targets from semantically similar background regions, especially under low-contrast or dynamic background conditions. Recent progress in Transformer-based architectures has provided new opportunities for global spatiotemporal modeling [33]. Building upon this, we propose a semantic affinity-driven spatiotemporal transformer network, which explicitly learns pixel-level semantic consistency between foreground objects and background elements. This approach significantly improves segmentation robustness and accuracy in challenging satellite video scenarios.

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III. Semantic Affinity-Driven Spatiotemporal Transformer Network

Both unsupervised and semi-supervised state-of-the-art methods for VOS exhibit significant limitations. On the one hand, unsupervised techniques often rely on complex postprocessing steps, which hinder their practicality and scalability. On the other hand, although semi-supervised approaches incorporate temporal information, their use of spatiotemporal attention modules still falls short in fully capturing the intricate relationships across both spatial and temporal dimensions. This limitation becomes particularly evident in large-scale satellite video scenarios, where numerous background elements closely resemble the object of interest. In such cases, prior methods tend to misclassify these similar regions as background, thereby limiting the network's ability to learn robust and generalized semantic representations. Therefore, to address these challenges, it is essential to effectively leverage both temporal consistency and spatial detail. The proposed method introduces a novel semantic affinity-driven framework that better models spatiotemporal dependencies. The details of our approach are described below.

A. Overall Architecture

Fig. 3 schematically illustrates our three-stage architecture: 1) a feature extraction module incorporating shape priors; 2) a global spatiotemporal semantic relationship modeling module; and 3) a contextual semantic affinity-constrained mask decoding module. In the satellite video single object segmentation process, the model initially receives three inputs: the image frame to be segmented, a reference set consisting of an image and its corresponding mask, and a reference edge mask. Each input is individually processed through several convolutional layers, and the resulting features are concatenated. Subsequently, the network fuses these features through extensive contextual interactions to further enhance edge salience. Finally, the flattened feature representation is fed into an encoder-decoder based Transformer network, which effectively models spatiotemporal relationships.

Given a satellite video sequence $X = \{X_1, X_2, \ldots, X_i, \ldots\}$ and an initial segmentation mask M_1 that delineates the initial position and shape of the object of interest, our objective is to generate a corresponding set of segmented outputs $Y = \{Y_1, Y_2, \ldots, Y_i, \ldots\}$, where each Y_i represents the segmentation of frame X_i.

Considering the unique characteristics of satellite video data and the spatial distribution of objects within these scenes, our method is specifically designed to achieve efficient object segmentation and robust semantic information learning. This approach provides essential support for applications such as maritime monitoring.

B. Feature Extraction Module Incorporating Shape Prior

To fully leverage the spatiotemporal information in both the reference sets and the query frame, as well as effectively integrate object-specific prior information, a robust feature extractor is crucial. This extractor captures object features and projects them into an embedding space for subsequent processing by the Vision Transformer. In our proposed method, a unified feature extractor is employed to process both the query frame and the reference sets.

Specifically, we design a triple-branch input layer that adaptively supports three types of input configurations: 1) query RGB frames alone, without any additional guidance; 2) reference RGB frames paired with object masks, used to provide semantic guidance for target identification; and 3) query RGB frames paired with edge masks derived from the object masks, which incorporate shape priors and serve as auxiliary boundary cues. This design allows the model to flexibly handle different levels of supervision during inference while maintaining a unified architecture. The edge masks are generated by applying contour extraction methods (e.g., Canny or Sobel) to the object masks in the reference frame, ensuring spatial alignment and semantic consistency between the edge mask and the original annotation. As illustrated in Fig. 3, the query frame is processed through the first branch using a standard convolution operation. For the reference set, three parallel convolutional operations are applied to encode the RGB frame, the foreground of the object mask, and the background, respectively. The edge mask is obtained by binarizing the object mask annotation from the reference frame and applying a boundary-tracking method to delineate the contours of all nonzero pixels, thereby extracting precise boundary points and hierarchical structural information. This edge mask maintains spatial and semantic consistency with the object mask in the reference set, ensuring that the derived contour representation accurately aligns with the target's shape in the original annotation. This edge mask is then fed into a standard convolution operation to enhance the learning and fusion of fine-grained details. The outputs from these three branches are subsequently concatenated to form the final feature representation.

This design allows for the flexible integration of any convolutional network as the feature extractor by replacing the initial layer with the proposed triple-branch input layer. In this work, we adopt the first four stages of the ResNet [34] as our feature extractor. After processing through the triple-branch input layer, the features from the query frame and the reference sets are concatenated along the temporal dimension before being fed into the subsequent CNN.

C. Spatiotemporal Semantic Modeling Module

Our approach fully leverages the extensive spatiotemporal redundancy in satellite video data by explicitly modeling both local and global semantic relationships. In the spatial domain, two rounds of cross-context semantic learning combined with a CNN-based feature extractor capture fine-grained interpixel relationships, producing precise segmentation masks [35]. In the temporal domain, a Transformer encoder computes attention along the spatiotemporal dimensions to enforce semantic consistency across frames, extracting robust and coherent target features [33]. This integrated design effectively distinguishes moving objects of interest from static background interference and accommodates temporal variations, thereby enhancing segmentation accuracy in complex scenes. The detail is shown in Fig. 4.

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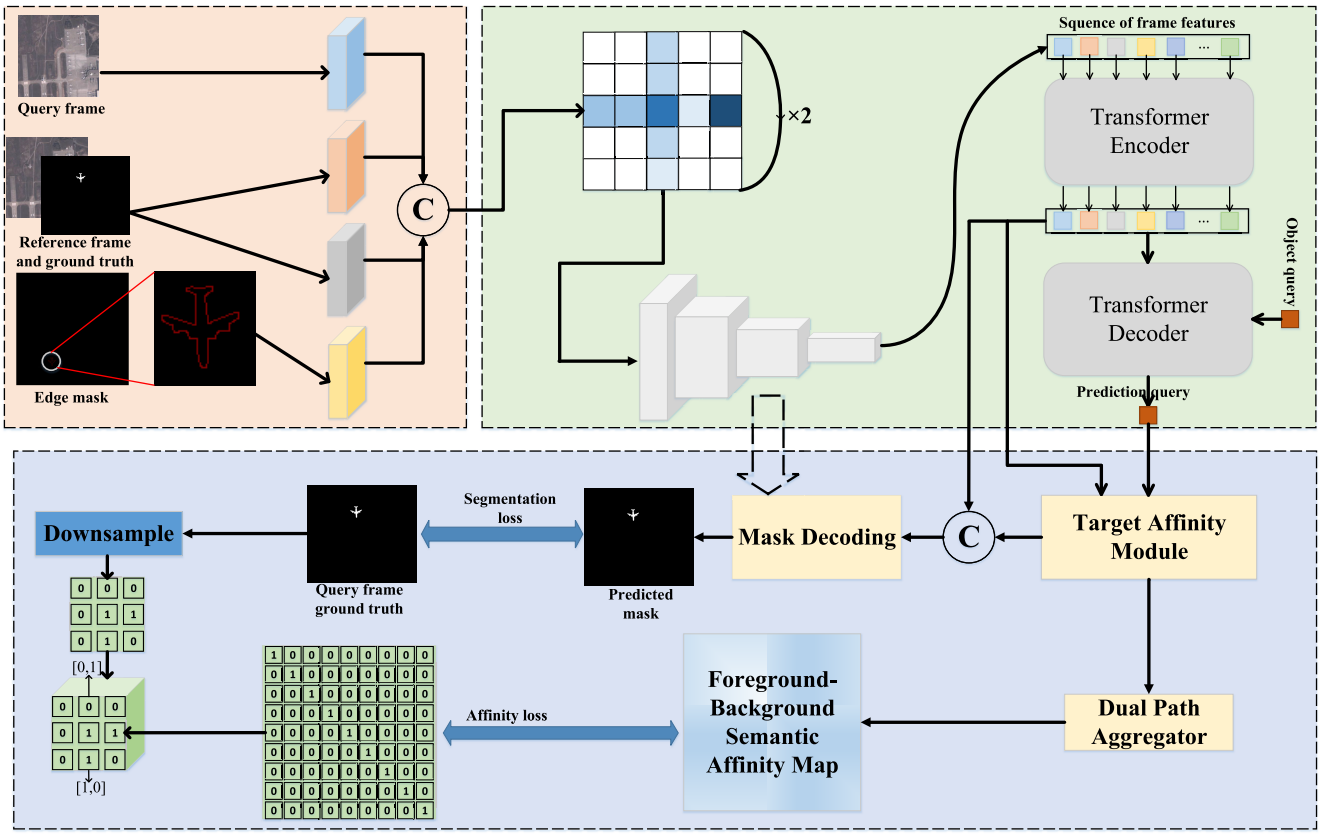


Fig. 3. General framework of the proposed semantic affinity-driven spatiotemporal Transformer network for moving-object segmentation in satellite videos.

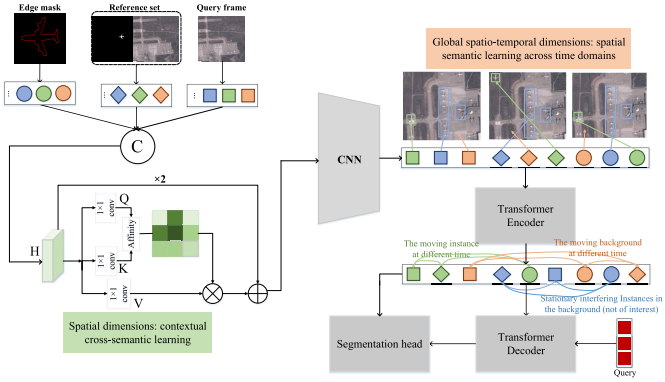


Fig. 4. General framework of the spatiotemporal semantic modeling module.

Denote the final output of the feature extraction module as H. It is followed by three 1×1 convolutional layers producing feature maps Q, K, and V. Here, $Q, K \in \mathbb{R}^{C' \times W \times H}$ with $C' < C$ and V maintains the same shape as H.

Each spatial location u in Q corresponds to a vector $Q_u \in \mathbb{R}^{C'}$; the same position in K gives $K_u \in \mathbb{R}^{C'}$. For each u, we collect K_u vectors along the horizontal and vertical axes, forming $(H + W - 1)$ semantic points (excluding overlap). This yields a matrix $\mathbb{R}^{(H+W-1)\times C'}$.

We compute dot-product affinities by multiplying $Q_u \in \mathbb{R}^{C'}$ with the transposed $(H + W - 1) \times C'$ matrix from K, resulting in a $(H+W-1) \times 1$ attention vector. As this is done for every spatial location, we obtain $W \times H$ such vectors. Stacking them yields a feature map $D \in \mathbb{R}^{(H+W-1)\times W \times H}$.

This operation enables strong spatial feature interaction, enhancing stability and accuracy in low-level representation. The resulting maps are then passed through the first four stages of ResNet. A 7×7 convolution and 3×3 max-pooling reduce resolution while retaining essential features. Subsequent residual stages incorporate 3×3 convolutions, batch normalization, and ReLU, gradually reducing spatial size and enriching semantics. Residual connections ensure gradient flow and efficient training.

Finally, the Transformer encoder is employed to capture spatiotemporal dependencies among pixel-level features in the sequence. It accepts an input feature set, denoted as Z, and generates encoded features E. The encoder comprises $N = 6$ layers, each built with a standard architecture that includes a multihead self-attention mechanism followed by a fully connected feed-forward network. This attention module is designed to extract spatiotemporal relationships across various representational subspaces. Specifically, let an element $z_{p,t} \in \mathbb{R}^d$ represent a feature at spatial position p and temporal position t from Z. For each attention head m (with $m \leq 8$), we derive the query, key, and value vectors from $z_{p,t}$. These vectors are represented as $q^m_{p,t}$, $k^m_{p,t}$, and $v^m_{p,t}$, respectively,

$$q^m_{p,t} = W^m_q z_{p,t}, \quad k^m_{p,t} = W^m_k z_{p,t}, \quad v^m_{p,t} = W^m_v z_{p,t}. \tag{1}$$

The multihead self-attention output, denoted by $\hat{z}_{p,t}$, is computed as follows. For each attention head m, the attention mechanism aggregates information from the entire reference

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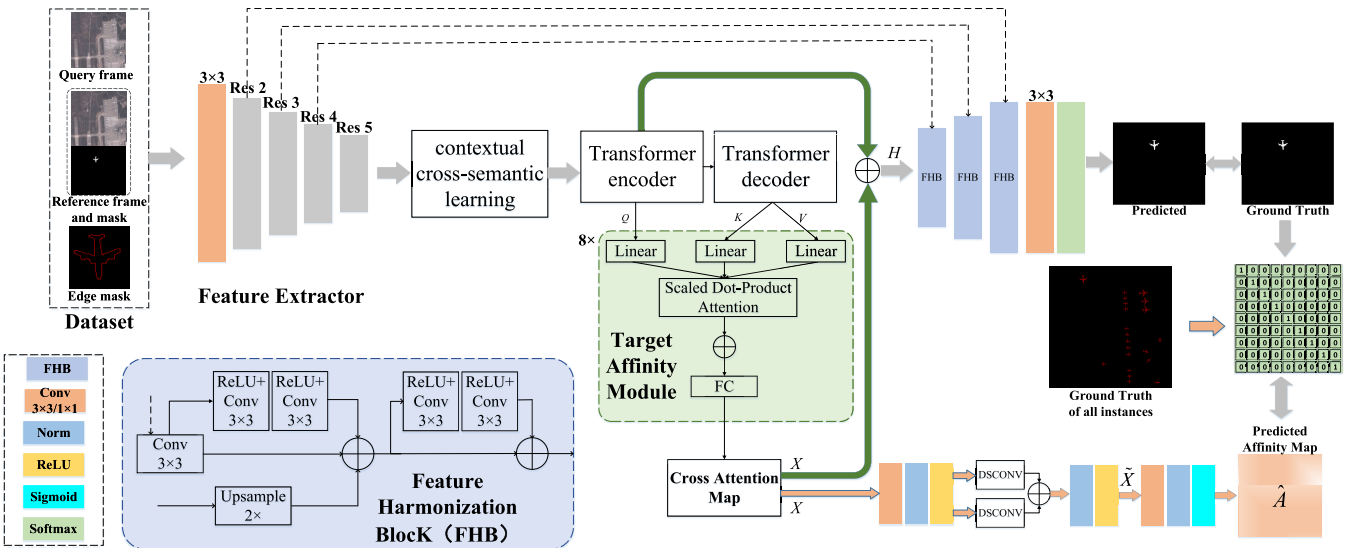


Fig. 5. Detail architecture of the segmentation with contextual semantic affinity constraint.

set. Specifically, we have

$$\hat{z}_{p,t} = \sum_{m=1}^{M} W_o^m \left[\sum_{t=1}^{T+1} \sum_{p=1}^{HW} \sigma \left(\frac{\left(q_{p,t}^m\right)^T k_{p,t}^m}{\sqrt{d_m}} \right) v_{p,t}^m \right] \tag{2}$$

where T represents the size of the reference set, and W_q^m, W_k^m, and W_v^m are learnable weight matrices in $\mathbb{R}^{d_m \times d}$. Moreover, W_o^m is a learnable weight in $\mathbb{R}^{d \times d_m}$, with d_m typically set to $d/8$. The function σ denotes the Softmax, which normalizes the attention scores. Importantly, attention is computed along the spatiotemporal axis, enabling simultaneous modeling of both spatial and temporal relationships.

D. Contextual Semantic Affinity-Constrained Mask Decoding Module

Contextual dependencies play a critical role in scene understanding and are particularly important in satellite VOS. In complex scenes such as airports and ports, stationary background objects (e.g., airplanes of different models) may share high-level semantic categories with moving targets but differ in appearance. Treating them uniformly as background can introduce bias and degrade performance.

To address this, our method introduces category-aware affinity constraints that model fine-grained semantic variations between foreground and background objects, enhancing segmentation precision and reducing misclassification due to semantic similarity.

Fig. 5 illustrates the architecture of our segmentation head with contextual semantic affinity constraint. To predict the target mask from the Transformer outputs, we first extract mask features from the query frame. The encoder feature E and decoder output feature D are then fed into a multihead self-attention module, where E acts as the query and D serves as both the key and value. This allows each attention head to compute the similarity between E and D, generating cross-attention maps that enhance feature interaction and fusion.

The resulting M attention maps, together with E, are concatenated into a feature volume H, which is used for final mask generation. Unlike simple concatenation or summation strategies, our design enables dynamic modeling of foreground–background relationships through attention mechanisms, leading to improved segmentation accuracy

$$\mathrm{Att}_i(D, E) = \sigma \left(\frac{\left(W_q^i D\right)^T \left(W_k^i E\right)}{\sqrt{d_i}} \right). \tag{3}$$

Here, i indexes the attention heads and $W_q^i, W_k^i \in \mathbb{R}^{d_i \times d}$ are learnable weight matrices, with $d_i = d/M$ by default.

The segmentation branch fuses the concatenated features H with the feature extractor's features via skip-connections to produce a mask. The feature harmonization block is used for efficient multiscale feature merging. The segmentation head comprises three such blocks, followed by a convolution and a Softmax layer to produce an object mask. Each block outputs a 256-channel feature map, while the final layer generates a two-channel mask map [36].

In the affinity prediction branch [37], the output cross attention map $X \in \mathbb{R}^{H \times W \times C_0}$ is first adapted to $\tilde{X} \in \mathbb{R}^{H \times W \times C_1}$ via a feature fusion module. A subsequent 1×1 convolution, batch normalization, and sigmoid activation are then applied to $\tilde{X}$ to produce the predicted affinity map $\hat{A} \in \mathbb{R}^{H \times W \times N}$ (with $N = H \times W$). To effectively capture local spatial information for semantic correlation, the feature fusion module utilizes fully separable convolutions. Specifically, a standard $k \times k$ convolution is factorized into a $k \times 1$ convolution followed by a $1 \times k$ convolution.

Next, we construct the supervisory label information for contextual semantic affinity constraints. Using the ground truth of a given input, we can determine the "semantic affinity" for each pixel, distinguishing pixels that belong to the same category from those that do not. This semantic affinity serves as a guiding principle for the network's learning process, rooted in the information provided by the ground truth. To achieve this, we construct an ideal affinity map from the ground truth to act

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as a supervision. For an input image I and its corresponding ground truth G, the image I is passed through the network to produce a feature map X with dimensions $H \times W$.

To align the ground truth G with the size of the feature map X, we first downsample G, resulting in a reduced ground truth G_e. Using a hot one-hot encoding, we convert the categorical labels in G_e into a matrix G_b of size $H \times W \times C$, where C represents the number of categories. The encoded ground truth is then reshaped into a matrix of size $N \times C$, where $N = H \times W$. By multiplying the matrix $A = G_b G_b^\top$, we obtain the ideal affinity map A, a matrix of size $N \times N$ that captures the relationship of categories between pixels. This ideal affinity map is used to guide the learning of the affinity map.

Each pixel in the affinity map represents a binary classification task. A typical approach to solving this involves applying binary cross-entropy loss. Considering the predicted affinity map P with dimensions $N \times N$, where $\{p_n \in P, n \in [1, N^2]\}$, and the reference ideal affinity map A, where $\{a_n \in A, n \in [1, N^2]\}$, the binary cross-entropy loss can be formulated as

$$L_{\text{aff}} = -\frac{1}{N^2}\sum_{n=1}^{N^2}(a_n \log p_n + (1-a_n)\log(1-p_n)). \tag{4}$$

For segmentation prediction output branch, we add cross-entropy loss L_{cls} and mask intersection over union (IoU) loss L_{IoU} as the single-object training loss L_{seg}. The segmentation loss can be expressed as

$$L_{\text{seg}} = L_{\text{cls}}(M_m, Y_m) + L_{\text{IoU}}(M_m, Y_m) \tag{5}$$

$$L_{\text{cls}}(M_m, Y_m) = -\frac{1}{|\Omega|}\sum_{p\in\Omega}\sum_{j=0}^{O-1}\left[Y_m \log\left(\frac{\exp(M_m)}{\sum_{j=0}^{O-1}\exp(M_j)}\right)\right]_p \tag{6}$$

$$L_{\text{IoU}}(M_m, Y_m) = 1 - \frac{\sum_{p\in\Omega}\min\left(Y_m^p, M_m^p\right)}{\sum_{p\in\Omega}\max\left(Y_m^p, M_m^p\right)}. \tag{7}$$

L_{seg} denotes the segmentation loss of the network, L_{cls} represents the pixel-wise classification loss, L_{IoU} denotes the mask prediction loss, and L_{aff} represents the affinity loss. Here, Ω refers to the set of all pixels in the motion target mask, M_m and Y_m represent the predicted and ground-truth masks for the motion target m, respectively. O is the number of motion targets. N is the size of the ideal affinity graph mentioned in the affinity constraint, and n represents the feature points in the feature map, with a_n being the predicted probability output by the network, and p_n denoting the true label, representing the target probability distribution.

Finally, the total loss of the network is calculated as

$$L_{\text{total}} = L_{\text{aff}} + L_{\text{seg}}. \tag{8}$$

IV. Experiment and Analysis

A. Dataset

The satellite video data employed in our experiments is sourced from SAT-MTB [38]. As shown in Table I, the dataset comprises 30 video segments from diverse scenes, totaling 5951 frames. The training set includes 22 video segments with 4460 frames, while the test set comprises eight video segments with 1491 frames. The videos have a spatial resolution of approximately 1 m and a frame rate of 10 frames/s, with each scene featuring a typical moving object. Fig. 6 illustrates example frames from two distinct scenes.

TABLE I
Dataset Statistics for Train ing and Testing Splits

dataset	train		test		total
	airplane	ship	airplane	ship	
number of videos	9	13	6	2	30
number of frames	1827	2633	140	626	5226

The data of satellite video depicting scenes of airplanes at land-based airports and ships on the sea, as illustrated in the sample images. Fig. 7 shows the data labels at different levels of granularity. At the object level, coarse-grained annotation labels all stationary background objects with a uniform semantic color. In contrast, instance-level fine-grained annotation assigns distinct semantic colors to each stationary background object, which are then used to construct different ideal affinity maps.

B. Implementation Details

We use the parameters of the model pretrained on the DAVIS dataset [39] for initializing the model. For training, we randomly sample $T = 5$ frames from a video and apply data augmentations to create a training video clip. This approach enables our model to capture long-range spatiotemporal dependencies effectively. During the online reference phase, our model segments the current frame by utilizing previous frames along with their predicted masks. Instead of storing the features of all past frames in an external memory, we only use the first frame with ground truth and the prior frames with their predicted masks as reference sets. The initial frame provides the most reliable information, while the subsequent frames are the closest in appearance to the current one, ensuring the model leverages the most relevant context for segmentation.

The model is configured with six layers for both the Transformer encoder and decoder. The multihead self-attention mechanism employs eight heads, each with a feature dimension of 256. The feed-forward networks consist of 2048 hidden units, and a dropout rate of 0.1 is applied to mitigate overfitting. Input images are standardized to a resolution of (480, 854), and each training video clip has a length of 2. Batch normalization layers are frozen during training. We use the AdamW optimizer to minimize the loss, with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay set to 10^{-4}. The learning rate starts at 10^{-4}, and training is conducted with a batch size of 2 over 160 epochs. Our experiments are conducted on a high-performance workstation equipped with four NVIDIA GeForce RTX 3090 GPUs, each featuring 24 GB of memory.

C. Evaluation Metrics

J-Mean, F-Mean, and J&F Mean are commonly used metrics in image segmentation tasks. In this study, these metrics

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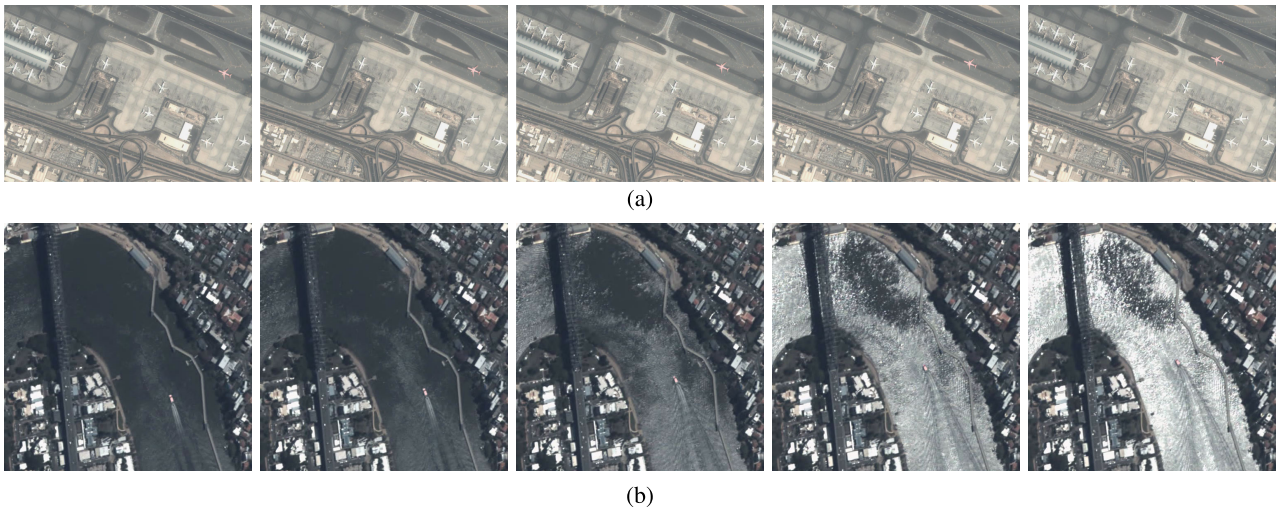


Fig. 6. Example frames of the dataset. (a) Airplane scene. (b) Ship scene.

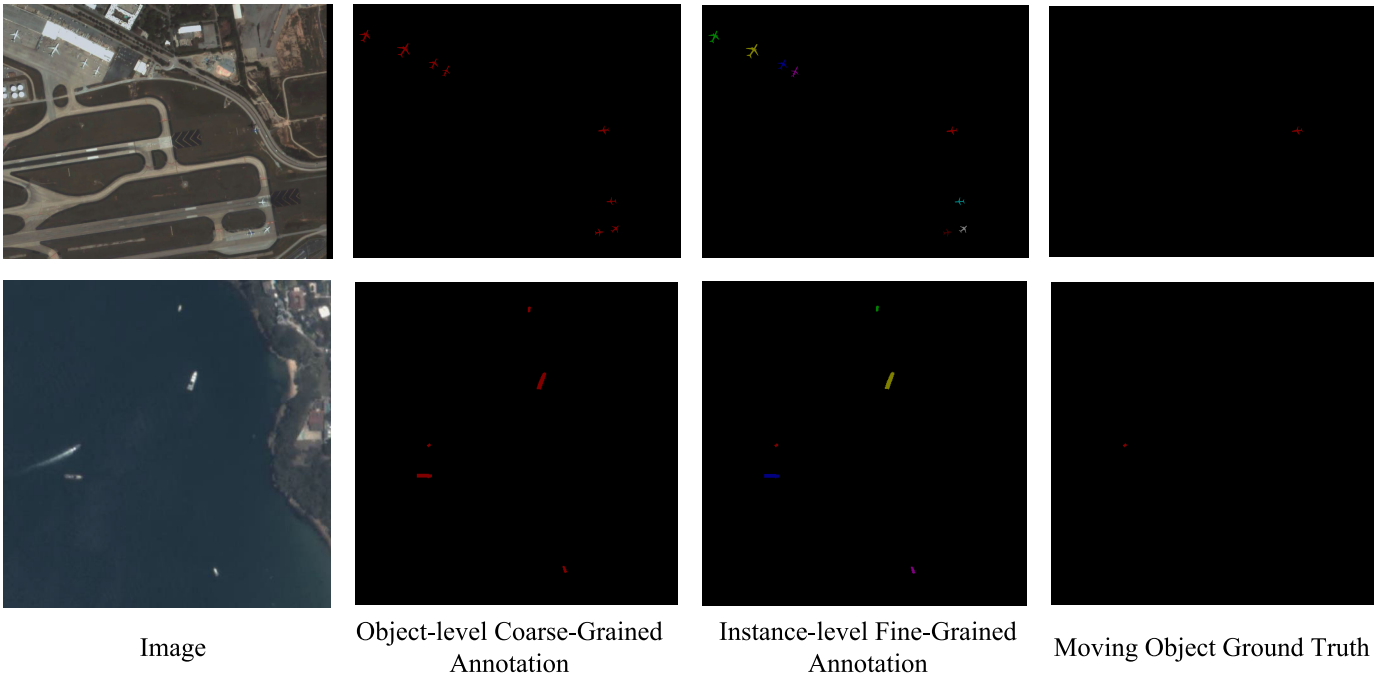


Fig. 7. Data labels at different levels of granularity.

are employed to evaluate the similarity between segmentation results and ground-truth labels. They are widely adopted in tasks such as VOS.

J-Mean represents the average IoU between the segmentation results and ground-truth labels. It ranges from 0 to 1 and is calculated by averaging the IoU values across all samples. This metric measures the degree of overlap between the predicted and ground-truth regions. J is calculated as follows:

$$J = \frac{|P \cap G|}{|P \cup G|} \tag{9}$$

where P represents the predicted mask, G denotes the ground-truth mask, $|P \cap G|$ is the number of pixels in the intersection of P and G, and $|P \cup G|$ is the number of pixels in their union.

F-Mean is the average $F1$-score based on boundary precision, assessing the alignment between the segmentation results and ground-truth boundaries. It is also bounded between 0 and 1. The specific calculation is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$F_1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \tag{12}$$

J&F Mean is a simple arithmetic mean of J-Mean and F-Mean, providing a comprehensive evaluation of both the segmentation region and boundary performance. By integrating region overlap (J-Mean) and boundary alignment

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(F-Mean), J&F Mean serves as a more holistic metric for assessing segmentation quality, especially in tasks where boundary details are critical

$$J\&F\text{Mean} = \frac{J\text{-Mean} + F\text{-Mean}}{2}. \tag{13}$$

D. Ablation Study

1) Ablation Experiments for the Proposed Modules: To comprehensively evaluate the effectiveness of the proposed modules, we conduct an ablation study by incrementally integrating each component into our spatiotemporal semantic learning networks. The experimental results are summarized in Tables II–IV.

Table II presents a comprehensive ablation study comparing different module configurations for satellite VOS. The baseline method achieves a J&F Mean of 0.680, a J-Mean of 0.443, and an F-Mean of 0.916. When the affinity constraint is introduced, the scores improve to 0.709, 0.482, and 0.935, respectively, while the shape prior module results in 0.702, 0.471, and 0.933. The context modeling module yields scores of 0.707, 0.476, and 0.938. Notably, when combining affinity constraint with context modeling or with shape prior, slight improvements are observed, with the former achieving a J&F Mean of 0.707, a J-Mean of 0.474, and an F-Mean of 0.940, and the latter reaching a J&F Mean of 0.710, a J-Mean of 0.484, and an F-Mean of 0.937. Ultimately, our proposed approach, which integrates all key components, attains the highest performance with a J&F Mean of 0.717, a J-Mean of 0.487, and an F-Mean of 0.947. This corresponds to an improvement of approximately 3.7% in the J&F Mean, 4.4% in the J-Mean, and 3.1% in the F-Mean over the baseline. These results clearly demonstrate that our method, through effective incorporation of affinity-guided spatiotemporal learning, shape-aware segmentation, and contextual modeling, robustly captures both global and local segmentation cues and significantly enhances performance in dynamic satellite video scenarios.

For qualitative evaluation, we visualize segmentation results on two test videos: one featuring an airplane target and another featuring a ship target. Specifically, we present the 100th frame from the airplane video and the 150th frame from the ship video. These frames occur at relatively late stages of the respective sequences, where segmentation performance typically begins to degrade. Nevertheless, the results demonstrate that our approach exhibits only minimal degradation over time. As shown in Fig. 8, the airplane's left wing is still clearly segmented, whereas the baseline method fails to capture it. For the ship target, the baseline approach erroneously includes a large area of the wake, whereas our method effectively reduces the mis-segmentation of the wake, resulting in a smaller and more accurate mask.

The loss curves for the ablation experiment are shown in Fig. 9. Experimental results show that incorporating shape priors and affinity constraints greatly stabilizes convergence and lowers loss compared to the baseline model. This enhanced model benefits from improved segmentation accuracy by leveraging both global and local contextual information, effectively suppressing noise and promoting robust feature learning.

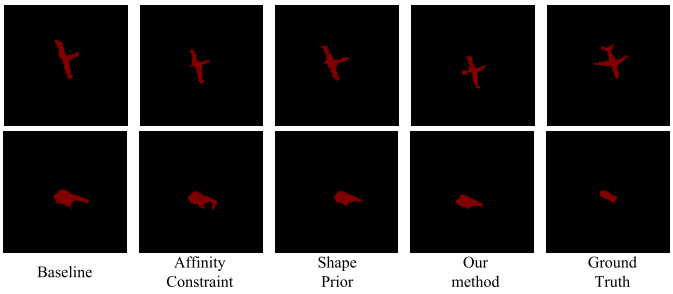


Fig. 8. Comparison of visualization results of modular ablation experiments.

TABLE II
ABLATION STUDY WITH COMPREHENSIVE METRICS ACROSS FULL TEST SETS (VALUES ARE UNITLESS, HIGHER IS BETTER)

Module	J&F Mean	J-Mean	F-Mean
Baseline	0.680	0.443	0.916
Affinity constraint	0.709	0.482	0.935
Shape prior	0.702	0.471	0.933
Context modeling	0.707	0.476	0.938
Shape prior Context modeling	0.704	0.472	0.937
Affinity constraint Context modeling	0.707	0.474	0.940
Affinity constraint Shape prior	0.710	0.484	0.937
Ours	**0.717**	**0.487**	**0.947**

TABLE III
COMPARISON OF INDICATORS OF DIFFERENT METHODS ON THE TOTAL TEST SET

Methods	J&F Mean	J-Mean	F-Mean
GMM	0.457	0.260	0.655
LBP	0.007	0.001	0.012
Frame-Difference	0.250	0.063	0.437
Optical-Flow	0.012	0.014	0.011
OSVOS	0.434	0.296	0.572
Transductive VOS	0.318	0.296	0.339
EMVOS	0.419	0.228	0.703
Optimized SAM2	0.655	0.380	0.929
Ours	**0.717**	**0.487**	**0.947**

Overall, the integration of these constraints significantly boosts performance and reliability.

The ablation study clearly demonstrates that the proposed affinity constraint and shape prior significantly enhance both temporal coherence and spatial accuracy. Additionally, the synergy between these modules surpasses their standalone effects, substantiating the efficacy of our holistic design.

2) Ablation Experiments for Different Fine-Grained Levels of the Affinity Constraint Module: We conduct exploratory experiments on fine-grained affinity constraints between the foreground and the background. We argue that the affinity between moving foreground objects and similar stationary background objects can be defined along two dimensions. The first dimension is category-level affinity: all objects belong to the broader category of aircraft and share similar structural

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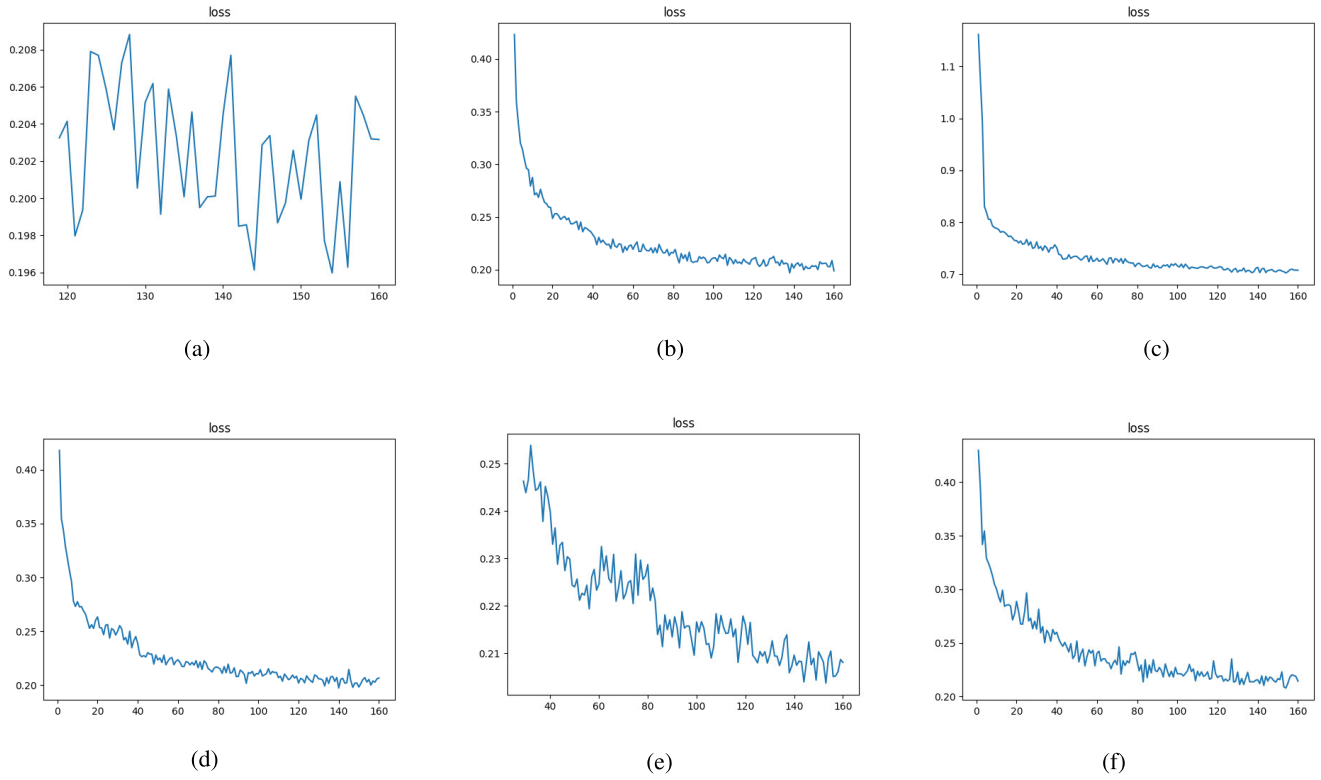


Fig. 9. Loss curve. (a) Baseline. (b) Baseline + shape prior. (c) Baseline + affinity constraint. (d) Baseline + shape prior + context modeling. (e) Baseline + context modeling + affinity constraint. (f) Proposed model.

TABLE IV
EXPERIMENTAL RESULTS OF ABLATION STUDY ON VIDEO TEST SEQUENCES

Baseline	Affinity constraint	Shape Prior	Context Modeling		Ship 1	Ship 2	Air1	Air 2	Air 3	Air 4	Air 5	Air 6
✓	-	-	-	J-Mean	0.255	0.474	0.594	0.197	0.446	0.605	0.221	0.751
				F-Mean	0.688	0.933	0.824	0.994	1	1	0.889	1
✓	✓	-	-	J-Mean	0.254	0.488	0.789	0.264	0.449	**0.627**	0.234	0.748
				F-Mean	0.687	0.910	1	0.998	1	1	0.888	1
✓	-	✓	-	J-Mean	0.251	0.395	0.782	0.256	**0.464**	0.617	0.240	0.764
				F-Mean	0.686	0.897	1	0.995	1	1	0.889	1
✓	-	-	✓	J-Mean	0.251	0.460	0.781	0.266	0.440	0.613	0.232	0.768
				F-Mean	0.703	0.905	1	0.997	1	1	0.895	1
✓	-	✓	✓	J-Mean	0.274	0.363	0.787	0.264	0.452	0.618	**0.240**	**0.778**
				F-Mean	0.692	0.925	1	0.995	1	1	0.883	1
✓	✓	-	✓	J-Mean	0.272	0.460	0.780	0.262	0.432	0.600	0.226	0.758
				F-Mean	0.723	0.905	1	0.998	1	1	**0.891**	1
✓	✓	✓	-	J-Mean	0.239	**0.496**	**0.789**	**0.279**	0.448	0.611	0.239	0.770
				F-Mean	0.655	**0.950**	1	**1**	1	1	0.890	1
✓	✓	✓	✓	J-Mean	**0.416**	0.438	0.788	0.254	0.426	0.594	0.225	0.753
				F-Mean	**0.847**	0.849	**1**	**0.995**	**1**	**1**	0.885	**1**

TABLE V
COMPARSION OF EXPERIMENTAL RESULTS WITH DIFFERENT PARTICLE SIZE LABELS ON EACH TEST VIDEO SEGMENT

Algorithm		Ship 9	Ship 35	Air 9	Air 59	Air 56	Air 47	Air 39	Air 25
Coarse-Grained Annotation	J-Mean	0.088	0.157	0.212	0.264	0.398	0.234	0.397	0.329
	F-Mean	0.194	0.757	0.436	0.818	0.702	0.769	0.853	0.711
Fine-Grained Annotation	J-Mean	0.260	0.321	0.786	0.246	0.450	0.613	0.227	0.766
	F-Mean	0.701	0.903	1	0.996	1	1	0.881	1

TABLE VI
COMPARSION OF EXPERIMENTAL RESULTS WITH DIFFERENT PARTICLE SIZE LABELS ON THE TOTAL TEST SET

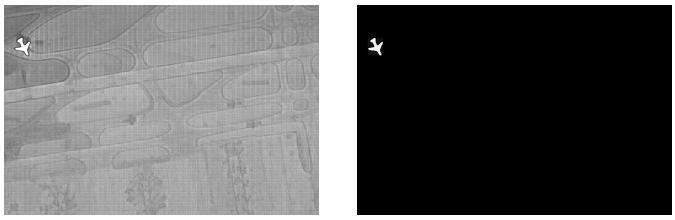
Methods	J&F Mean	J-Mean	F-Mean
Coarse-grained Annotation	**0.709**	**0.482**	0.935
Fine-grained Annotation	0.697	0.459	0.935

components (e.g., nose, wings, tail, engine, etc.). The second dimension is model-level fine-grained matching: for example, if the moving object is a Boeing 737, a stationary Boeing 737 in the background will exhibit the highest affinity, whereas other aircraft will show varying degrees of affinity based on subtle structural differences.

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TABLE VII
COMPARISON OF INDICATORS OF DIFFERENT METHODS ON EACH TEST VIDEO SEGMENT

Algorithm		Ship 9	Ship 35	Air 9	Air 59	Air 56	Air 47	Air 39	Air 25
GMM [40]	J-Mean	0.088	0.157	0.212	0.264	0.398	0.234	0.397	0.329
	F-Mean	0.194	0.757	0.436	0.818	0.702	0.769	0.853	0.711
LBP [41]	J-Mean	0	0	0	0	0	0	0.001	0.001
	F-Mean	0.013	0.028	0.016	0.004	0.003	0.012	0.010	0.013
frame-difference [42]	J-Mean	0.008	0	0.033	0	0.013	0.111	0.323	0.015
	F-Mean	0.075	0.002	0.562	0	0.659	0.811	0.909	0.477
optical-flow [43]	J-Mean	0.082	0.022	0.001	0	0	0.001	0.004	0.002
	F-Mean	0.080	0	0.001	0	0	0	0.002	0
OSVOS [5]	J-Mean	0.287	0.305	0.302	0.070	0.164	0.425	**0.398**	0.417
	F-Mean	0.574	0.580	0.527	0.313	0.390	0.713	0.749	0.727
Transductive VOS [44]	J-Mean	0.250	0.310	0.280	0.320	0.290	0.305	0.315	0.300
	F-Mean	0.340	0.350	0.330	0.360	0.345	0.355	0.320	0.310
EMVOS [45]	J-Mean	0.3	0.03	0.197	0.002	0.54	0.292	0.171	0.297
	F-Mean	0.786	0.528	0.456	0.038	0.938	0.839	0.562	0.733
Optimized SAM2 [46]	J-Mean	**0.710**	**0.531**	0.222	**0.268**	0.395	0.296	0.254	0.367
	F-Mean	0.840	**0.951**	0.879	**0.996**	1	0.916	0.879	0.974
Ours	J-Mean	0.416	0.438	**0.788**	0.254	**0.426**	**0.594**	0.225	**0.753**
	F-Mean	**0.847**	0.849	**1**	0.995	**1**	**1**	**0.885**	**1**



(a) (b)

Fig. 10. Comparison of SAM2 results before and after optimization. (a) Original model. (Contrast adjusted for visual analysis.) (b) Bbox prompt for each frame and ROI postprocessing.

TABLE VIII
ANALYSIS OF THE EXPERIMENTAL SETUP OF SAM2

Settings	J&F Mean	J-Mean	F-Mean
thresholding	0.421	0.001	0.840
box prompt	0.489	0.001	0.976
box prompt+ROI	**0.655**	**0.380**	**0.929**
Ours	**0.717**	**0.487**	**0.947**

Tables V and VI present a comparison between two different affinity label annotation strategies. The results indicate that while both approaches yield improvements over the baseline, the coarse-grained annotation achieves slightly better performance. Specifically, the coarse-grained method attains a J&F Mean of 0.709 and a J-Mean of 0.482, whereas the fine-grained method results in a J&F Mean of 0.697 and a J-Mean of 0.459, with the F-Mean remaining constant at 0.935 for both. This reflects a 2.3% drop in the J-Mean metric when using fine-grained labels, suggesting that the increased granularity does not translate into enhanced spatial accuracy and may even detract from performance. The drop in J-Mean when using fine-grained labels may be attributed to the limited dataset size. With insufficient data, the network struggles to learn the subtle differences and matching relationships among various target models, which undermines its ability to accurately delineate object boundaries, resulting in lower J-Mean scores.

E. Comparison Experiment

1) Analysis of Experimental Results of Different Methods: In the comparative experiments, our method is evaluated against four traditional unsupervised techniques (GMM, LBP, frame-difference, and optical-flow), four deep learning-based semi-supervised VOS methods (OSVOS, EMVOS, RVOS, and transductive VOS), as well as the currently popular foundational visual model SAM2 (segment anything model 2). SAM2 is developed by OpenAI and represents a cutting-edge segmentation framework that leverages large-scale pretraining on diverse visual datasets, enabling robust and precise segmentation across various scenarios. Tables VII and III provide a detailed comparison of various VOS methods, evaluated using the J&F Mean, J-Mean, and F-Mean metrics. Figs. 11 and 12 present the visualization results of these methods in different scenes, where each group corresponds to one model. The two figures, respectively, depict the third frame and the penultimate frame from a test video sequence, providing an intuitive illustration of the methods' performance degradation

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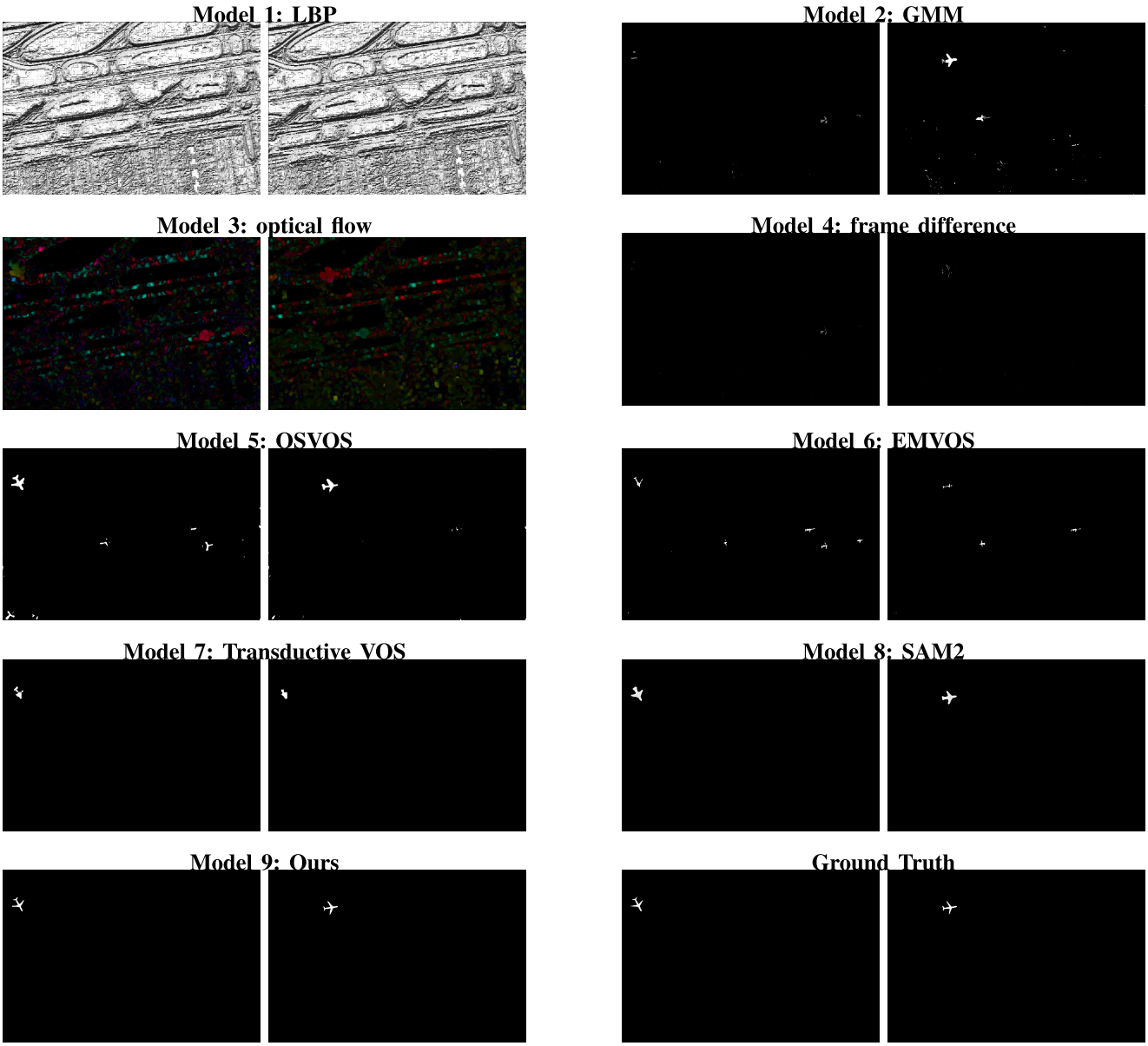


Fig. 11. Visualization of segmentation results of different methods in aircraft target scenarios.

over time. Finally, by leveraging the ground-truth bounding box to zero. Fig. 10 shows the comparison of SAM2 results before and after optimization.

Traditional unsupervised approaches, such as GMM, LBP, frame difference, and optical flow, exhibit relatively low performance across these metrics. Notably, LBP and optical flow methods perform particularly poorly, with LBP achieving a J&F Mean of 0.007 and optical flow registering a J&F Mean of 0.012. These results highlight the limitations of traditional methods in capturing complex object boundaries and temporal dynamics. As shown in Figs. 11 and 12, these approaches are susceptible to background motion, illumination shifts, and other environmental variations inherent in satellite imagery. Consequently, they often produce substantial noise and spurious artifacts, leading to the inclusion of irrelevant regions in the segmented results. These findings highlight the limitations of unsupervised segmentation methods for dynamic satellite scenes, where the overall camera motion and frequent intensity fluctuations can easily be misinterpreted as foreground objects.

In contrast, deep learning-based semi-supervised methods demonstrate improved segmentation performance. OSVOS achieves a J&F Mean of 0.434, with a J-Mean of 0.296 and an F-Mean of 0.572, indicating moderate segmentation accuracy. EMVOS shows similar performance, with a J&F Mean of 0.419, a J-Mean of 0.228, and an F-Mean of 0.703. OSVOS and EMVOS exhibit notable segmentation deficiencies in satellite video scenes. Specifically, multiple stationary background objects are erroneously included as foreground, while the aircraft's fuselage and wings are fragmented or partially missing. Furthermore, as time progresses, the segmentation quality degrades substantially, indicating that these models struggle to maintain consistency across longer sequences. The optimized SAM2 model, a current state-of-the-art visual foundation model, achieves a high F-Mean of 0.929 but a

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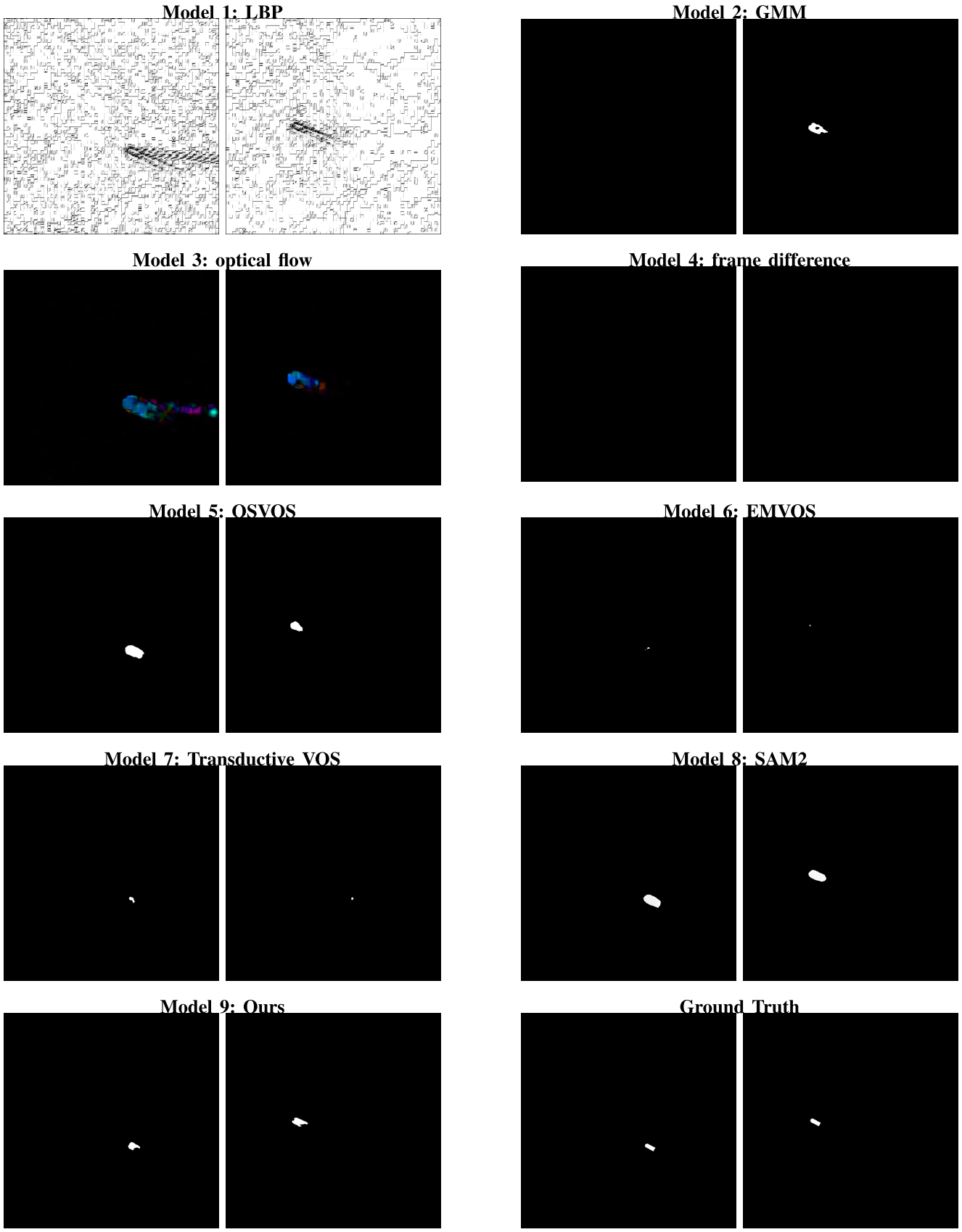


Fig. 12. Visualization of segmentation results of different methods in ship target scenarios.

low J-Mean of 0.380, suggesting strong boundary accuracy but poor spatial segmentation consistency.

Our proposed method significantly outperforms these baselines, achieving a J&F Mean of 0.717, a J-Mean of 0.487, and

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an F-Mean of 0.947. Compared to OSVOS, this represents an approximate 65% improvement in both J-Mean and F-Mean scores. These substantial enhancements underscore the superior capability of our approach in robustly capturing both global and local segmentation cues. Consequently, our method is particularly effective in addressing the demands of dynamic satellite video scenarios, where accurate delineation of moving objects is critical.

2) Optimization of the Analysis of SAM2 Results: In our experiments, we utilize the SAM2.1_hiera_large model, a state-of-the-art segmentation network that features a hierarchical Transformer backbone. This large variant is distinguished by its increased parameter count relative to its medium and small counterparts, enabling it to capture multiscale contextual features more effectively. The backbone's design, which integrates multilevel attention and refined patch embedding mechanisms, facilitates superior feature extraction from complex scenes.

When segmenting satellite video data, SAM2 tends to erroneously extract background textures and other noisy pixels beyond the intended target. Moreover, the segmentation boundaries are often overly smooth and exhibit a slight outward expansion. While this approach effectively covers the target region, it compromises the precision and detail of the contour delineation, particularly for objects like aircraft.

In the first set of experiments, we used the original default settings. For each test video, only the mask label for the initial frame was provided; however, the segmentation results revealed that SAM2 erroneously segmented noise pixels, such as the textured airport background, leading to a J-Mean score that was nearly zero. In the second set of experiments, we applied a threshold to eliminate low-intensity background noise, but the outcome remained unsatisfactory. In the third set of experiments, we incorporated the ground-truth bounding box labels as constraints during the inference of each video frame, which yielded a slight improvement, although the J-Mean score still remained at a low level. Finally, by leveraging the ground-truth bounding box regions, we forcibly retained the mask within the bounding box for each frame while directly setting the background to zero.

As shown in Table VIII, even when SAM2 is provided with ample prior information and optimized via postprocessing, its segmentation results do not exceed those of our proposed method. Furthermore, our model contains approximately 34.6 million parameters, compared to 224 million in SAM2, while achieving competitive performance. This leads to significant advantages in both model compactness and deployment efficiency, making it more suitable for resource-constrained satellite video applications.

V. Conclusion

In this article, we presented a semantic affinity-driven spatiotemporal Transformer network for accurate moving-object segmentation in satellite videos. Our method systematically addresses three core challenges in this field: spatiotemporal redundancy, semantic ambiguity between moving and stationary objects, and weak feature discrimination under low-resolution imaging. By leveraging a Transformer-based architecture to capture fine-grained pixel dependencies across space and time, integrating a contextual affinity-constrained decoder to suppress category-level interference, and enhancing contour representation through edge-aware feature extraction, the proposed framework achieves superior segmentation performance.

Extensive experiments conducted on a dataset derived from SAT-MTB demonstrate that our approach not only surpasses existing baselines in terms of J&F metrics but also achieves significant improvements in model efficiency. Without requiring additional fine-tuning during inference, our framework exhibits strong generalization capability and deployment practicality.

In future work, we plan to extend this framework to more complex satellite video analysis tasks, including multiobject tracking and scene-level event understanding. Moreover, further research into adaptive temporal aggregation and multimodal data fusion could enhance robustness under more diverse observation conditions. We believe that the proposed method provides an effective and scalable solution for intelligent satellite video interpretation, contributing to broader advancements in Earth observation technologies.

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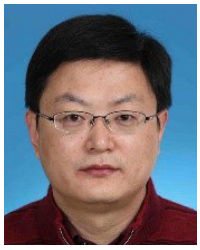
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